Urinary Bladder Pathology — II

Pathology GUE Block · MEDucation by Mattin Majid

Part II of this SGL — continued from Part I: renal tumors.

BEFORE YOU READ — ANSWER FROM MEMORY

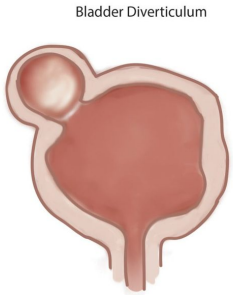
1. Which type of bladder diverticulum is more common, and what are its three clinical consequences?
2. How does WHO 2022 distinguish papilloma, PUNLMP, low-grade, and high-grade papillary urothelial carcinoma by urothelial thickness and cytologic atypia?
3. What is the AJCC/UICC T-stage for lamina propria invasion vs. muscularis propria invasion in bladder carcinoma?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 Congenital anomalies

Diverticula

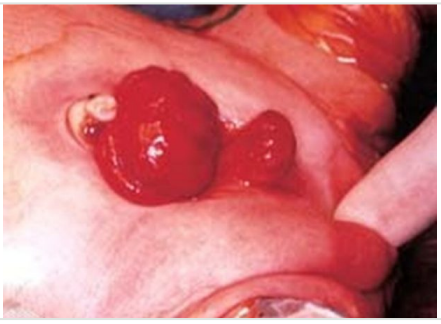
What	Round to ovoid sac-like pouch. May be congenital, but more commonly **acquired**.
Congenital	Due to failure of development of normal musculature, or urinary tract obstruction during fetal development.
Acquired	Due to **prostatic enlargement** (hyperplasia or neoplasia).
Clinical significance	Most are small and asymptomatic. May cause: **urinary stasis** (predisposes to infection and bladder calculi); **vesicoureteral reflux** (impingement on ureter); rarely, **carcinomas may arise** in bladder diverticula.



Bladder diverticulum

Exstrophy

What	A developmental failure in the anterior wall of the abdomen and the bladder; the bladder communicates directly with the abdominal surface.
Consequence	The exposed bladder mucosa develops infections, and may undergo **colonic glandular metaplasia**.
Risk	Exstrophy is associated with an increased risk of **bladder adenocarcinoma** later.



Bladder exstrophy

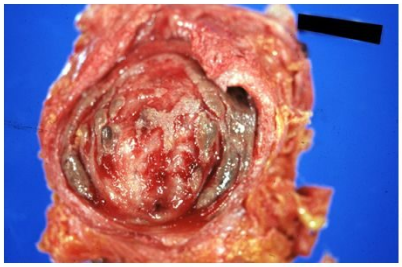
Urachal anomalies

What	The urachus is a fibrous remnant of the allantois (a canal that drains the urinary bladder of the fetus and runs within the umbilical cord), normally disappears before birth.
Patency	May remain patent — total or partial. **Total patency:** fistula between bladder and umbilicus. **Partial patency:** urachal cysts lined by transitional or metaplastic epithelium; carcinomas (mostly **adenocarcinoma**) may arise in such cysts.

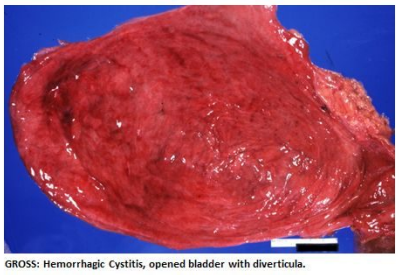
2 Inflammation

Acute and chronic cystitis

WHO	Women are more likely to develop cystitis because of their shorter urethra.
Etiologic agents	Bacteria (*E. coli*, *Proteus*), TB cystitis, fungi (*Monilia*), schistosomiasis, viruses (adenovirus), chlamydia, and mycoplasma.
Predisposing factors	Bladder calculi and urinary obstruction, diabetes mellitus, instrumentation, immune deficiency, chemotherapy, and radiation.
Clinically	Triad of: **frequency, suprapubic pain, and dysuria.**
Morphology	Hyperemia of mucosa. Exudate: suppurative (purulent) cystitis. Hemorrhage: hemorrhagic cystitis. Cystitis following radiation or chemotherapy: epithelial atypia. **Chronic cystitis** differs from acute in the character of the inflammatory infiltrate and fibrous thickening of the bladder wall.



Cystitis — ulcerative purulent mucosal surface

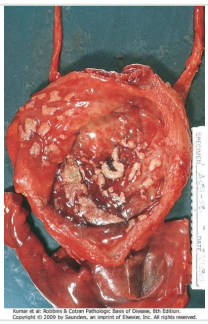


Hemorrhagic cystitis — opened bladder with diverticula

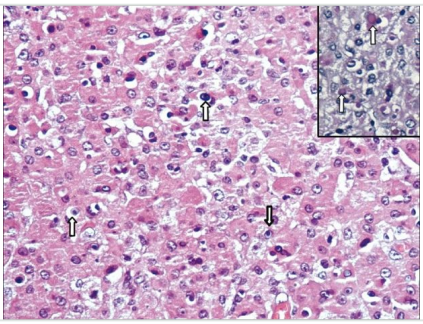
Special forms of cystitis

Chronic pelvic pain syndrome / Interstitial cystitis (Hunner ulcer)	Pain — pressure & discomfort; occurs most frequently in **women**; inflammation and fibrosis of **all layers** of bladder wall. Cystoscopy: fissures and punctate hemorrhages (glomerulations) in bladder mucosa which later ulcerate. Main role of biopsy is to **rule out CIS**. Etiology unknown.
Polypoid cystitis	Irritation of bladder mucosa, most commonly as a result of instrumentation (catheter). **May be mistaken for papillary urothelial carcinoma.**

Malakoplakia — chronic inflammation; soft yellowish mucosal plaques 3–4 cm; related to chronic bacterial infection (*E. coli* or *Proteus*), frequent in immunosuppressed or transplant recipients; due to defects in phagocytic/degradative function of macrophages that become overloaded with undigested bacterial products. Microscopy: infiltration of bladder mucosa by large macrophages with abundant granular cytoplasm; laminated mineralized concretions from calcium deposition (**Michaelis-Gutmann bodies**), present both within macrophages and in between. Similar lesions can occur in colon, lung, bone, kidney, prostate, and epididymis.



Malakoplakia (gross) — soft, yellow mucosal plaques



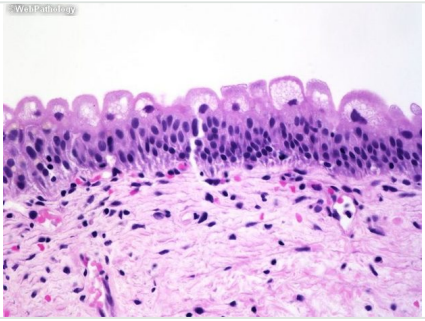
Numerous intracellular and extracellular Michaelis-Gutmann bodies (arrows)

3 Urothelial tumors — WHO classification 2022

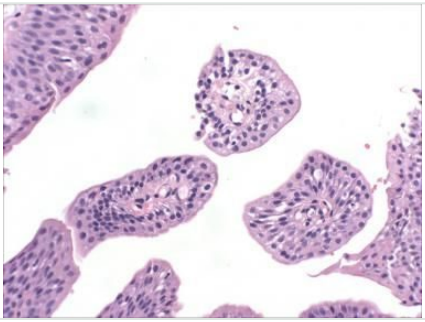
Noninvasive urothelial tumors: urothelial papilloma; inverted urothelial papilloma; papillary urothelial neoplasm of low malignant potential (PUNLMP); noninvasive papillary urothelial carcinoma, low grade; noninvasive papillary urothelial carcinoma, high grade; urothelial carcinoma in situ.

Invasive urothelial neoplasms: invasive urothelial carcinoma.

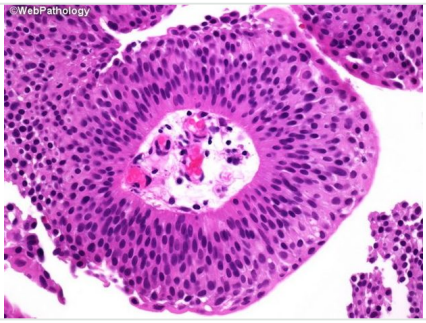
Lesion	Key features
Papilloma	Benign tumor, commonly seen in younger patients. Finger-like papillae with fibrovascular core covered by transitional epithelial cells identical to normal urothelium (**7–8 cell layer** thickness). Umbrella cell layer present.
PUNLMP	Only difference from papilloma is thicker urothelium (**>8 cell layers**). Mitotic figures not seen (when present, confined to basal layer). Not associated with invasion. Umbrella cell layer may be present or absent.
Low-grade papillary UC	Thickness is more, but tumor cells still maintain polarity (cohesive), and show mild–moderate atypia (scattered hyperchromatic nuclei, infrequent mitoses, mild–moderate pleomorphism). Loss of umbrella cell layer. **Can recur and can invade.**
High-grade papillary UC	Tumor cells discohesive with loss of polarity, severe atypia (large hyperchromatic nuclei, frequent mitoses, marked pleomorphism, necrosis). Loss of umbrella cell layer. Higher incidence of invasion into muscular layer and significant metastatic potential.
Carcinoma in situ (CIS)	Flat lesion — presence of malignant cells in **flat urothelium**, not invading the basement membrane. Range from full-thickness to scattered malignant cells (pagetoid spread).



Papilloma — fibrovascular core covered by urothelial cells



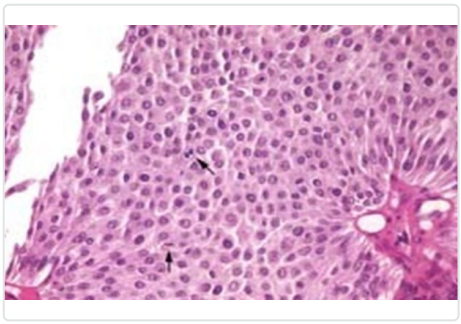
Papilloma — multiple papillae



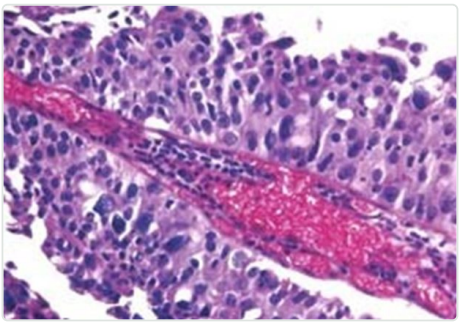
PUNLMP



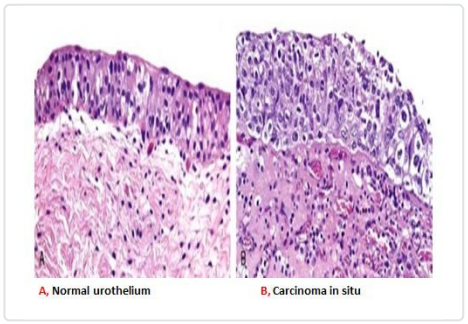
PUNLMP — papillary fronds



Low-grade papillary urothelial carcinoma



High-grade papillary urothelial carcinoma



A. Normal urothelium B. Carcinoma in situ

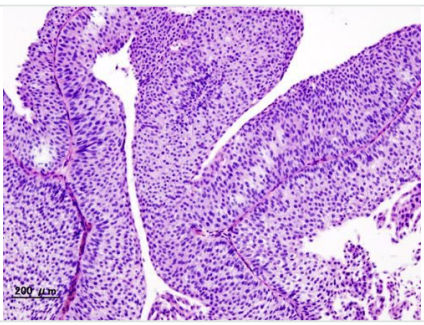
Invasive urothelial (transitional cell) cancer — extent of invasion (flat or papillary) is of prognostic significance. Staging (in addition to grading) is critical in assessment of bladder neoplasms.

AJCC/UICC pathologic staging of bladder carcinoma (T = primary tumor)

Depth of invasion	Stage
Noninvasive, papillary	**Ta**
Carcinoma in situ (noninvasive, flat)	**Tis**
Lamina propria invasion	**T1**
Muscularis propria invasion	**T2**
Microscopic extravesical invasion	**T3a**
Grossly apparent extravesical invasion	**T3b**
Invades adjacent structures	**T4**

Epidemiology, clinical, and pathogenesis of urothelial (transitional cell) cancer

WHO	More common in **males** (ratio 3:1).
Presentation	Most often: **hematuria.**
Risk factors	Cigarette smoking; industrial exposure to naphthylamine (an aniline dye); long-term use of analgesics; long-term exposure to cyclophosphamide; prior exposure of bladder to radiation.



Invasive papillary urothelial carcinoma

4 Other types of bladder carcinomas

Squamous cell carcinoma	More frequent in countries endemic for **urinary schistosomiasis**; can also be associated with renal calculi.
Adenocarcinoma	Arise from **urachal remnants**, or **exstrophy**.

5 Mesenchymal tumors

Benign	Rare; the most common is **leiomyoma**.
Malignant (sarcomas)	Most common sarcoma in **infancy/childhood**: **embryonal rhabdomyosarcoma**. Most common sarcoma in **adults**: **leiomyosarcoma**.

P Must-remember pearls

- Bladder diverticula are more commonly **acquired** (prostatic enlargement) than congenital, and predispose to urinary stasis, calculi, vesicoureteral reflux, and rarely carcinoma.
- **Exstrophy** and **urachal remnants** both predispose to bladder **adenocarcinoma** — exstrophy via colonic glandular metaplasia of exposed mucosa, urachal remnants via patent urachal cysts.
- Cystitis triad: **frequency, suprapubic pain, dysuria.** Interstitial cystitis (Hunner ulcer) biopsy is mainly done to **rule out CIS**.
- **Malakoplakia** shows **Michaelis-Gutmann bodies** (laminated calcium concretions) within macrophages overloaded with undigested bacterial debris — same lesion can occur in colon, lung, bone, kidney, prostate, epididymis.
- WHO 2022 grading ladder by urothelial thickness/atypia: papilloma (7–8 layers, normal) → PUNLMP (>8 layers, no atypia) → low-grade (mild–moderate atypia, cohesive) → high-grade (severe atypia, discohesive, necrosis) → CIS (flat, malignant cells, no invasion).
- Bladder cancer: male predominance 3:1, hematuria most common presentation; risk factors — smoking, naphthylamine dye exposure, analgesic abuse, cyclophosphamide, radiation.
- SCC of bladder is linked to **urinary schistosomiasis**; adenocarcinoma arises from **urachal remnants or exstrophy**.
- Pediatric bladder sarcoma = **embryonal rhabdomyosarcoma**; adult bladder sarcoma = **leiomyosarcoma**.

Q Test yourself

1. A biopsy from a bladder lesion shows urothelium 10 cell layers thick, cells maintaining polarity, with mild scattered hyperchromatic nuclei and infrequent mitoses. The umbrella cell layer is lost. This best fits:

A. Urothelial papilloma
B. PUNLMP
C. Low-grade papillary urothelial carcinoma
D. High-grade papillary urothelial carcinoma

2. A patient with a history of prostatic hyperplasia develops a bladder outpouching. Which complication of this lesion is specifically due to impingement on the ureter?

A. Urinary stasis
B. Vesicoureteral reflux
C. Bladder calculi
D. Carcinoma arising in the pouch

3. A renal transplant recipient develops soft yellow bladder mucosal plaques. Biopsy shows macrophages with granular cytoplasm and laminated calcified concretions. What are these concretions called?

A. Psammoma bodies
B. Michaelis-Gutmann bodies
C. Call-Exner bodies
D. Councilman bodies

4. Bladder biopsy shows tumor invading into the muscularis propria but not beyond the bladder wall. What AJCC/UICC T-stage is this?

A. T1
B. T2
C. T3a
D. T3b

5. A patient from a region endemic for urinary schistosomiasis develops a bladder malignancy. Which histologic subtype is most strongly associated with this risk factor?

A. Urothelial carcinoma
B. Squamous cell carcinoma
C. Adenocarcinoma
D. Leiomyosarcoma

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 · C — Thickened urothelium with preserved cell polarity, mild-moderate atypia, infrequent mitoses, and loss of umbrella cell layer defines low-grade papillary urothelial carcinoma.

2 · B — Vesicoureteral reflux results from the diverticulum impinging on the ureter; urinary stasis and calculi relate to stagnant urine pooling in the pouch itself.

3 · B — Michaelis-Gutmann bodies are the laminated, calcium-containing concretions characteristic of malakoplakia.

4 · B — Invasion into the muscularis propria (detrusor muscle) is staged as T2.

5 · B — Squamous cell carcinoma of the bladder is more frequent in countries endemic for urinary schistosomiasis.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam